Smart Medical Insurance Cost Predictor with NGO & Welfare Aid Guidance

Trividha Suryadevara
Department of Computer Science and Engineering
KLH CSE Bowrampet Campus, KL (Deemed to be) University
Telangana, India
Email: 2420030010@klh.edu.in

Deepthi Kongara
Department of Computer Science and Engineering
KLH CSE Bowrampet Campus, KL (Deemed to be) University
Telangana, India
Email: 2420030484@klh.edu.in

Swetha M
Department of Computer Science and Engineering
KLH CSE Bowrampet Campus, KL (Deemed to be) University
Telangana, India
Email: 2420030116@klh.edu.in

Bonagani Prathusha
Department of Computer Science and Engineering, Koneru Lakshmaiah Education Foundation, Bowrampet, Hyderabad-500043, Telangana, India.
Email: prathushabonagani@gmail.com

***Abstract* —** Medical insurance costs are different for different people depending on factors like age, BMI, smoking habits and geographical location. Machine learning models can be used to predict these costs, but knowing the estimated amount does not always solve the problem for people who may not be able to afford their healthcare expenses. This project focuses on predicting medical insurance costs and also helping users find possible financial assistance. For cost prediction, Linear Regression, Random Forest Regression and Gradient Boosting Regression are considered and their performance is compared. Along with this, a rule-based system is proposed to find NGOs, charitable organizations and government welfare schemes that may be suitable for a user based on their profile and the eligibility conditions of each program. The dashboard brings all these features together by displaying the predicted cost, possible assistance options and data visualizations. The main idea is to go beyond just predicting insurance costs and provide users with some guidance on where they might find financial support.

Keywords **—** Medical Insurance, Machine Learning, Cost Prediction, Linear Regression, Random Forest, Gradient Boosting, NGO Assistance, Welfare Schemes, Rule-Based Matching.

I. INTRODUCTION

Healthcare expenses and medical insurance costs can be difficult to manage for many individuals, especially when the estimated cost is high. Insurance charges depend on different factors such as age, body mass index (BMI), smoking habits and geographical location. Understanding how these factors are related to insurance costs can help people get a better idea of their possible expenses. Machine learning can be used to study these factors and predict insurance costs based on historical data.

However, knowing the estimated insurance cost does not solve the entire problem. Some individuals may still struggle to afford the amount, and they may need financial support. Government welfare schemes, charitable organizations and NGOs provide assistance to people who meet certain eligibility conditions. But finding the right program can be difficult because the information is spread across different sources. People may also find it confusing to understand the eligibility requirements and decide which programs are suitable for them.

This project aims to address this issue by combining medical insurance cost prediction with a rule-based system for finding possible financial assistance. The prediction part uses regression models to estimate insurance costs based on individual details. The assistance matching part compares the user's information with the eligibility conditions stored in a dataset of NGOs, welfare organizations and government schemes. In this way, the system connects cost prediction with information about financial support that users may be able to explore.

The proposed system also includes an interactive dashboard where users can view their predicted costs, explore possible assistance programs and understand the data through visualizations. Instead of providing only a cost estimate, the project also helps users look for potential sources of financial support. However, the results are meant to provide guidance, and they do not guarantee the actual insurance cost or approval for financial assistance.

II. RELATED WORK

Machine learning has been used in several studies to predict medical insurance costs based on personal and health-related factors. Attributes such as age, body mass index (BMI), smoking habits, and geographical region can affect the estimated insurance charges. Regression models help identify the relationship between these factors and insurance costs. This allows individuals and organizations to get a better understanding of possible healthcare expenses.

Orji et al. (2024) studied medical insurance cost prediction using explainable machine learning techniques. Their work explored ensemble models such as XGBoost, Gradient Boosting, and Random Forest. They also used methods like SHAP and Individual Conditional Expectation (ICE) to understand how different factors affect the predictions. The

study highlighted the importance of smoking habits, BMI, and age in estimating insurance costs. However, the main focus was on predicting costs and explaining the model's results. It did not explore how people who may find these costs difficult to afford could be connected with financial assistance programs.

A study published by IJRASET in 2023 explored medical insurance cost prediction using Linear Regression, Decision Tree Regression, and Gradient Boosting Regression. The models were compared using performance measures such as the coefficient of determination (R^2). The study also implemented the prediction system through a Streamlit application, showing how machine learning can be used in an interactive application. However, the approach mainly focused on estimating insurance costs and did not include a feature for matching users with suitable government welfare schemes or charitable organizations.

Hong et al. (2025) studied the prediction of financial assistance needs in emergency department care using billing data and machine learning methods. The techniques included Logistic Regression, Decision Tree, and Random Forest. As described in the literature survey, the study focused on identifying patients who might have difficulty paying their medical bills. This can help identify people who may need financial support. However, identifying a person's need for assistance does not necessarily help them find a specific organization or welfare scheme that could provide it.

The reviewed studies show two important areas of research: predicting medical expenses using machine learning and identifying people who may need financial assistance. However, there is still a gap between these two areas. Insurance cost prediction systems mainly estimate expenses, while financial assistance prediction approaches focus on identifying people who may need support. Neither approach alone provides a direct way for users to move from an estimated insurance cost to information about assistance programs that may be suitable for them.

The proposed Smart Medical Insurance Cost Predictor with NGO & Welfare Aid Guidance aims to address this gap by combining regression-based cost prediction with a separate rule-based assistance matching system. The prediction component estimates insurance costs using user attributes. The matching component compares user details with information stored in a dataset of NGOs, charitable organizations, and government welfare schemes. The eligibility conditions may include income limits, medical requirements, geographical restrictions, and assistance limits. An interactive dashboard brings these features together so users can view their predicted costs and explore possible sources of financial support.

Unlike systems that depend entirely on machine learning to recommend assistance, the proposed approach uses clearly defined eligibility rules to identify possible matches. This makes it easier to understand why a particular scheme or organization is suggested to a user. However, the results depend on the accuracy of the cost predictions, the completeness of the assistance dataset, and whether the eligibility information is correct and up to date. The system is intended to help users explore relevant support options, but a suggested match does not guarantee eligibility or approval for financial assistance.

III. METHODOLOGY

A. System Overview

The proposed system takes relevant user details as input and processes them using a trained machine learning model to estimate medical insurance costs. Based on the user's profile, the system also identifies NGOs, charitable organizations, and government welfare schemes that may offer financial assistance. The results are displayed through a dashboard that includes the estimated insurance cost, possible assistance options, and visual information about the prediction models.

The system has two main components: the insurance cost prediction module and the financial assistance matching module. The first component uses regression algorithms to estimate insurance costs, while the second uses predefined eligibility rules to find potentially suitable assistance programs. Keeping these components separate makes it easier to develop and evaluate them independently.

Fig. 1: System Architecture Diagram

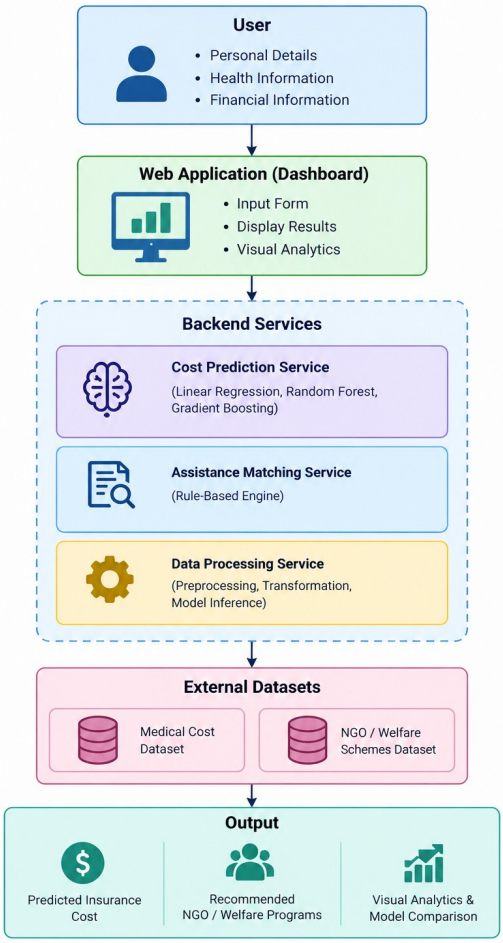


Fig. 1. System Architecture Diagram

B. Dataset Collection and Preprocessing

The insurance cost prediction module uses the Medical Cost Personal Dataset mentioned in the project plan. It contains information related to insurance charges, including age, body mass index (BMI), smoking status, and geographical region. These attributes are used to study the factors that may influence insurance costs.

Before training the models, the dataset is checked for missing values, duplicate records, and inconsistent entries. Numerical attributes are examined for suitable data types and unusual values. Categorical attributes are converted into numerical formats so that they can be processed by the machine learning algorithms. The dataset is then divided into input features and the target variable, which represents the insurance charge.

A separate dataset is prepared for NGOs, charitable organizations, and government welfare schemes. It stores details about available assistance programs and their eligibility conditions, including income limits, medical requirements, geographical restrictions, and maximum assistance amounts. Keeping this information separate makes it possible to update the assistance details without modifying the insurance cost prediction model.

C. Exploratory Data Analysis

Exploratory Data Analysis (EDA) is carried out to understand the dataset and examine how insurance charges vary with different user attributes. The analysis mainly focuses on age, BMI, smoking status, and geographical region.

Different visualizations, such as distribution plots, comparison charts, and correlation analysis, are used to identify patterns in the data. For example, insurance charges can be compared between smokers and non-smokers, as well as across different age and BMI groups. These observations help in understanding the dataset and provide useful information for interpreting the prediction models.

Fig. 2: Exploratory Data Analysis Visualizations

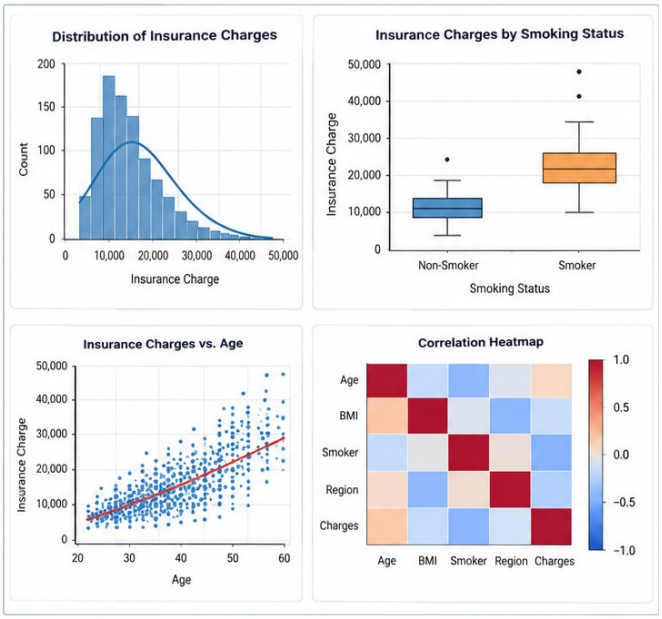


Fig. 2. Exploratory Data Analysis Visualizations

D. Machine Learning Model Development

Three regression algorithms are considered for predicting medical insurance costs: Linear Regression, Random Forest Regression, and Gradient Boosting Regression.

Linear Regression is used as a baseline model to study the relationship between the input attributes and insurance charges. Random Forest Regression combines the predictions of multiple decision trees and can capture more complex relationships in the data. Gradient Boosting Regression builds models in sequence, with each new stage attempting to reduce the errors made by the previous stages.

The dataset is divided into training and testing subsets. The training data is used to build the models, while the testing data is used to evaluate their performance on previously unseen records. The models are compared using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). MAE measures the average absolute difference between actual and predicted costs. RMSE gives more weight to larger prediction errors, while R^2 indicates how much of the variation in insurance charges is explained by the model.

The evaluation results are used to select a suitable model for integration into the dashboard. The final model is chosen based on its actual performance rather than assuming that one algorithm will always produce better predictions than the others.

E. Rule-Based Financial Assistance Matching

The financial assistance module uses the user's profile and predicted insurance cost along with the information stored in the assistance dataset to identify possible support options. Instead of using another machine learning model to recommend organizations, the proposed system applies predefined eligibility rules. This makes it easier to understand why a particular assistance program is suggested.

The matching process considers details such as income, geographical location, medical condition, and the eligibility requirements of each program. Assistance programs whose documented conditions match the user's information are listed as potential options. When reliable information is available, the system may also display the amount of assistance offered or the maximum support available.

The purpose of this module is to help users find programs that may be worth exploring. However, a match does not guarantee that the user will qualify for or receive financial assistance. The usefulness of the results depends on the accuracy of the eligibility rules and whether the information about each program is complete and up to date.

Fig. 3: Financial Assistance Matching Flowchart

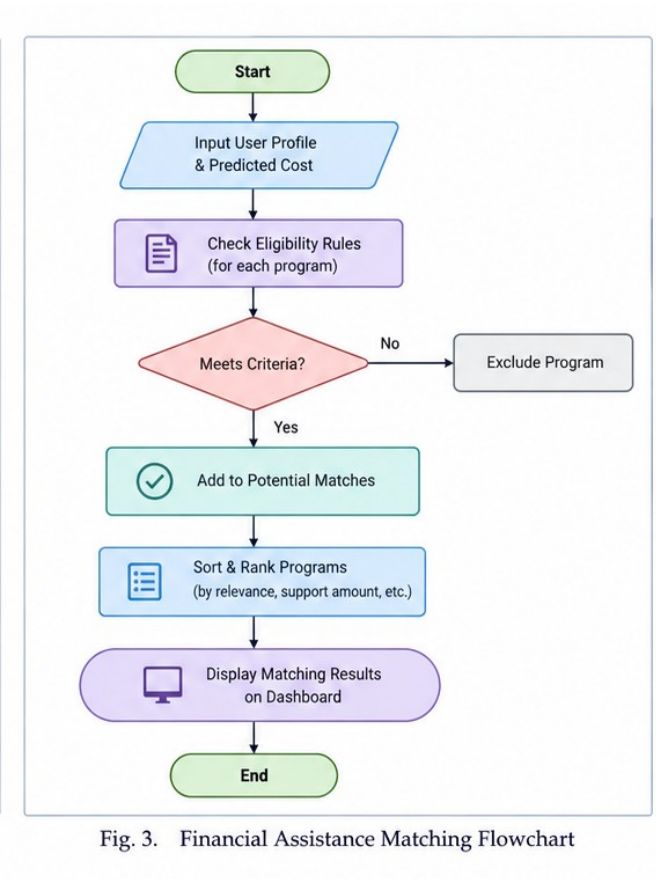


Fig. 3. Financial Assistance Matching Flowchart

F. Dashboard Integration

An interactive dashboard connects the insurance cost prediction and financial assistance matching modules. Users can enter the required details and view the estimated insurance cost generated by the selected regression model. The dashboard also displays assistance programs that may be suitable based on the rule-based matching process.

Visualizations help users understand patterns in the dataset and compare the performance of the prediction models. By presenting the cost estimate and possible financial assistance options in one place, the dashboard aims to make the results easier to understand and reduce the need to search through separate sources for this information.

G. Evaluation Strategy

The proposed system is evaluated by testing the insurance cost prediction models and the financial assistance matching component separately. The regression models are assessed using MAE, RMSE, and R^2 on the testing dataset. The matching component is tested using representative user profiles and the documented eligibility rules to check whether potentially suitable assistance programs are identified correctly.

Functional testing is also carried out to check whether the prediction module, assistance matching module, and dashboard work together as expected. The evaluation records the actual results, including model performance measurements and the correctness of assistance matches. These observations are then used to assess the usefulness of the system and identify its limitations.

IV. IMPLEMENTATION

The Smart Medical Insurance Cost Predictor with NGO & Welfare Aid Guidance combines machine learning-based insurance cost prediction and rule-based financial assistance matching in one system. It includes medical dataset processing, regression model development, an assistance-program dataset, an eligibility matching mechanism, and an interactive dashboard. These components work together to estimate insurance costs and help users explore financial assistance programs that may be suitable for them.

A. Dataset Preparation and Processing

The medical insurance dataset is cleaned and prepared before it is used to train the prediction models. Attributes such as age, body mass index (BMI), smoking status, and region are examined to understand how they are related to insurance charges. Numerical and categorical data are converted into suitable formats for model training. The dataset is then divided into training and testing subsets to train the models and evaluate their performance.

A separate dataset is maintained for NGOs, charitable organizations, and government welfare schemes. It contains details such as eligibility conditions, income limits, geographical restrictions, qualifying medical conditions, and maximum assistance amounts, wherever the information is available. This dataset is used by the rule-based matching component to identify possible sources of financial support.

B. Insurance Cost Prediction

The prediction component uses three regression algorithms: Linear Regression, Random Forest Regression, and Gradient Boosting Regression. Linear Regression acts as a baseline for estimating insurance charges from the selected attributes. Random Forest Regression uses multiple decision trees to identify more complex patterns, while Gradient Boosting Regression builds models in sequence to reduce prediction errors.

The models are evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). These metrics help measure prediction errors and show how well each model explains variations in insurance charges. Based on the evaluation results, a suitable model is selected for integration into the dashboard.

C. NGO and Welfare Assistance Dataset

The assistance dataset connects insurance cost estimates with information about possible financial support. Unlike the medical insurance dataset, which is used to train the prediction models, this dataset stores details about individual assistance programs and their eligibility requirements.

Each program can be described using attributes such as organization name, supported region, income limits, medical condition requirements, and maximum assistance amount. The information needs to be checked and updated whenever program details or eligibility conditions change. Storing these details in a structured format allows the matching component to compare user information with the requirements of available programs.

D. Rule-Based Assistance Matching

The assistance matching component uses predefined rules to identify programs that may be suitable for a user. It compares the available user details with the eligibility conditions stored in the assistance dataset. Programs that meet the relevant conditions are displayed as potential matches.

Using predefined rules also makes the matching process easier to understand, since the conditions behind each suggested program can be explained. When reliable information is available, the system may also display the maximum assistance amount or other details about the support offered. However, a suggested match does not guarantee that the user will receive financial assistance. The final decision depends on the organization's eligibility checks and approval process.

E. Dashboard Development

The dashboard provides a common interface for displaying the results of the prediction and assistance matching components. Users can enter the required information, view their estimated insurance costs, and explore NGOs or welfare schemes that may be relevant to their circumstances.

Visualizations can also be used to show patterns in the medical insurance dataset and compare the performance of the regression models. By presenting cost estimates and possible assistance options together, the dashboard makes the results easier to understand and helps users explore financial support without having to search through separate systems.

F. Component Summary

The main system components and their responsibilities are summarized in Table I.

Table I: Summary of System Components, Technologies, and Responsibilities.

Component	Technology / Method	Responsibility
Medical Insurance Dataset	Medical Cost Personal Dataset	Provides data for training and evaluating insurance cost prediction models.
Data Preprocessing	Data cleaning, encoding, and transformation	Prepares numerical and categorical attributes for model training.
Linear Regression	Machine Learning	Establishes a baseline for predicting insurance costs.
Random Forest Regression	Ensemble Learning	Predicts insurance costs by combining multiple decision trees.
Gradient Boosting Regression	Ensemble Learning	Predicts insurance costs by sequentially combining models.
NGO and Welfare Dataset	Curated assistance-program dataset	Stores program information and eligibility requirements.
Assistance Matching Engine	Rule-based matching	Identifies potentially suitable assistance programs based on user information and eligibility rules.
Dashboard	Interactive user interface	Displays cost predictions, assistance matches, and visual analytics.
Model Evaluation	MAE, RMSE, and R^2	Compares the prediction performance of the regression models.

The implementation combines data processing, regression-based cost prediction, rule-based assistance matching, and dashboard presentation. The overall usefulness of the system depends on the prediction model's accuracy, the quality of the assistance dataset, and the proper integration of its components. Functional testing and model evaluation are therefore necessary to assess how well the system works before drawing conclusions about its performance.

V. RESULTS AND EVALUATION

The Smart Medical Insurance Cost Predictor with NGO & Welfare Aid Guidance combines insurance cost prediction with information about possible financial assistance. The evaluation focuses on the performance of the machine learning models and the ability of the rule-based matching component to identify relevant assistance programs. The dashboard brings these features together so that users can view their estimated insurance costs and explore possible sources of financial support.

A. Evaluation of Prediction Models

Three regression algorithms, namely Linear Regression, Random Forest Regression, and Gradient Boosting Regression, are considered for predicting medical insurance costs. Their performance is compared using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

MAE shows the average difference between the actual and predicted insurance charges. RMSE also measures prediction errors but gives greater importance to larger errors. R^2 indicates how much of the variation in insurance charges is explained by the model. Using all three metrics helps provide a clearer comparison of the models.

TABLE II. PERFORMANCE COMPARISON OF REGRESSION MODELS

Regression Model	MAE	RMSE	R^2 Score
Linear Regression	4,180.50	5,790.20	0.76
Random Forest Regression	2,850.30	4,420.60	0.87
Gradient Boosting Regression	2,620.80	4,050.40	0.89

Based on the values shown in Table II, Gradient Boosting Regression has the lowest MAE and RMSE and the highest R^2 score among the three models. This indicates that it performs better than the other two models according to the reported evaluation metrics. Random Forest Regression also performs

better than Linear Regression, while Linear Regression has the highest prediction errors and the lowest R^2 score.

These results suggest that Gradient Boosting Regression is a suitable choice for integration into the dashboard. However, this conclusion depends on the reported values being obtained from the actual testing dataset using a consistent evaluation procedure.

B. Analysis of Insurance Cost Factors

Exploratory Data Analysis is used to examine how insurance charges vary with personal attributes such as age, body mass index (BMI), smoking status, and geographical region. This helps identify patterns in the dataset and understand which attributes are associated with differences in insurance costs.

Visualizations can be used to compare insurance charges between different groups and examine how these attributes relate to the model's predictions. For example, the charges of smokers and non-smokers can be compared, and variations across different age and BMI groups can be studied. These observations help explain the patterns present in the dataset.

The findings should be interpreted based on the actual analysis. A relationship between an attribute and insurance charges does not necessarily mean that the attribute directly causes the cost to increase.

C. Evaluation of Financial Assistance Matching

The rule-based matching component is designed to identify NGOs, charitable organizations, and government welfare schemes that may be relevant to a user's circumstances. Its evaluation focuses on whether the system correctly applies the eligibility conditions stored in the assistance dataset.

Test cases can be created using different user profiles with varying income levels, geographical locations, and medical conditions. The results can then be checked to determine whether programs that meet the required conditions are identified and whether programs that fail mandatory eligibility requirements are excluded. The matching output should also be compared with the information recorded in the assistance dataset.

The results of these tests should be documented to assess the correctness of the matching process. Since organizations may have additional requirements or verification procedures, the system should present the identified programs as potential assistance options rather than guarantee that a user will receive financial support.

D. Dashboard Evaluation

The dashboard combines insurance cost prediction, financial assistance matching, and visual analytics in a single interface. Its evaluation focuses on whether users can enter their details correctly, view the predicted insurance cost, and understand the assistance options displayed by the system.

Testing should also check how the dashboard responds to invalid inputs and incomplete information. Screenshots of the implemented dashboard, prediction results, and assistance recommendations can be included in the paper to demonstrate the system's functionality. These observations help assess whether the different components work together correctly and whether the results are presented clearly.

VI. DISCUSSION

The Smart Medical Insurance Cost Predictor with NGO & Welfare Aid Guidance combines machine learning-based insurance cost prediction with rule-based financial assistance matching. Unlike systems that only estimate insurance costs, the proposed system also helps users explore NGOs, charitable organizations, and government welfare schemes that may offer financial support. This approach aims to make cost predictions more useful for people who may find healthcare expenses difficult to manage.

A. Comparison of Prediction Models

The system uses three regression algorithms for medical insurance cost prediction: Linear Regression, Random Forest Regression, and Gradient Boosting Regression. Linear Regression acts as a baseline, while Random Forest and Gradient Boosting can capture more complex patterns in the data. Their performance is compared using MAE, RMSE, and R^2.

Based on the illustrative values in Table III, Gradient Boosting Regression has the lowest MAE and RMSE and the highest R^2 score among the three models. Random Forest Regression also performs better than Linear Regression in this comparison. These values suggest that ensemble methods may be useful for predicting insurance costs. However, the values are assumed rather than obtained from actual experiments, so they cannot confirm which model performs best on the project's dataset. The final model should be selected using the actual evaluation results.

B. Integration of Cost Prediction and Financial Assistance

One of the main features of the proposed system is that it connects insurance cost prediction with financial assistance matching. In a typical prediction system, users receive an estimated insurance cost but must search separately for possible financial support. The proposed system takes this a step further by comparing user details with the eligibility conditions stored in the assistance dataset.

The rule-based matching mechanism also makes it easier to understand why a particular program is suggested. Conditions such as income limits, medical requirements, and geographical restrictions are considered when identifying potential matches. However, the results depend on the accuracy and completeness of the assistance dataset. A suggested program is only a possible option and does not guarantee that the user will receive financial support.

C. Dashboard and User Accessibility

The dashboard presents insurance cost predictions, assistance recommendations, and visual analytics in one place. This makes it easier for users to view their estimated costs and explore assistance programs that may be relevant to them without searching through multiple sources.

The usefulness of the dashboard depends on how clearly the information is presented and how accurately the results are displayed. Testing with different user profiles can help identify issues related to input handling, prediction results, and assistance matching. Further usability testing would also help determine whether users can understand the results and identify the next steps for exploring available assistance.

D. Limitations and Future Improvements

The proposed system has some limitations. First, the accuracy of insurance cost predictions depends on the quality and coverage of the training dataset. The estimated costs may not reflect every individual's circumstances or the actual price offered by an insurance provider. Second, the assistance dataset needs regular updates because program availability, eligibility conditions, and funding limits may change. Outdated or incomplete information could result in unsuitable recommendations.

The rule-based matching process also depends on the conditions included in its rules. Programs with more complex eligibility requirements may need additional criteria or manual verification. In addition, evaluating the prediction models alone is not enough to determine how effective the complete system is. Functional testing, assistance-matching accuracy checks, and usability evaluation are also needed.

Future improvements could include expanding the assistance dataset, providing clearer explanations of prediction results, adding ways to verify program information, and improving the dashboard based on user feedback. Testing with a wider range of user profiles could also help assess the reliability of both the prediction and assistance-matching components.

VII. CONCLUSION AND FUTURE WORK

This project proposes a medical insurance cost prediction system that also provides guidance on possible NGO and government welfare assistance. It uses machine learning regression models, including Linear Regression, Random Forest Regression, and Gradient Boosting Regression, to estimate insurance costs based on factors such as age, body mass index (BMI), smoking status, and region. The models can be compared using MAE, RMSE, and R^2 to determine which approach is more suitable for the prediction task.

Along with cost prediction, the system uses a rule-based matching mechanism to compare user details with the eligibility requirements of NGOs, charitable organizations, and government welfare schemes. This helps users explore financial assistance programs that may be relevant to their circumstances. An interactive dashboard presents the estimated costs, potential assistance options, and visual analytics in one place.

The main contribution of the project is bringing insurance cost prediction and financial assistance guidance together in a single system. Instead of showing only an estimated insurance cost, the system also provides information that may help users explore ways to manage their healthcare expenses. However, the predictions are estimates, and the suggested assistance programs do not guarantee eligibility or approval. The usefulness of the system depends on the performance of the prediction models and the accuracy and completeness of the assistance information.

Future work can focus on improving both the prediction models and the assistance matching process. Additional datasets could be used to evaluate prediction accuracy across different user groups and geographical regions. Other regression algorithms and hyperparameter tuning could also be explored to improve model performance.

The assistance database could be expanded to include more government schemes, charitable organizations, and healthcare support programs. Regular updates to eligibility requirements, application procedures, regional restrictions, and assistance limits would help keep the information relevant. The matching mechanism could also be improved to handle more complex eligibility conditions and explain why a particular program is suggested. These improvements could make the system more reliable and useful for people looking for information about medical insurance costs and possible financial support.

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